

MI NIR API

Version 0.1

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REVISION HISTORY

Revision No.	Description	Date
0.1	Initial release	12/20/2021
0.2	Remove vis/contrast/saturation level	01/04/2022

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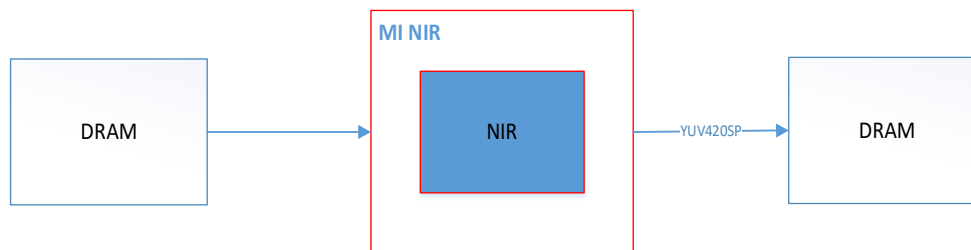
1. OVERVIEW

1.1. Module description

NIR Module is a 2-image fuser to combine a visible image and a near-infrared (NIR) image.

- It will extract the detail (high contrast) from both images
- It will reduce the noise level from both images

1.2. Flow chart



For input image format in dram, refer to the description of the following keywords.

1.3. Keyword

Input dram layout:



VIS:
Visible

NIR:
near-infrared

Device:
NIR Hardware device

Channel:
Time division multiplexed channel on device

Input image format:

YUV420SP

Output image format:

YUV420SP

Input resolution:

Muffin:

Maximum support 2688 x 1520

Support input image width 8Pixel alignment, Stride 16 alignment

Output resolution:

Same as input resolution

2. API Reference

This function module provides the following user APIs.

API Name	Function
MI_NIR_CreateDevice	Create a NIR Device
MI_NIR_DestroyDevice	Destroy a NIR Device
MI_NIR_CreateChannel	Create a NIR Channel
MI_NIR_DestroyChannel	Destroy a NIR Channel
MI_NIR_StartChannel	Start NIR Channel
MI_NIR_StopChannel	Stop NIR Channel
MI_NIR_SetChnParam	Set NIR Channel parameter
MI_NIR_GetChnParam	Get NIR Channel parameter

2.1. MI_NIR_CreateDevice

➤ Description

Create a NIR device.

➤ Syntax

MI_S32 MI_NIR_CreateDevice(MI_NIR_DEV DevId , [MI_NIR_DevAttr_t](#) *pstDevAttr);

➤ Parameter

Parameter name	Description	Input/Output
DevId	NIR Device ID	Input
pstDevAttr	NIR Device attribute pointer, static attribute	Input

➤ Return value

- MI_SUCCESS(0) successful.
- Non-Zero: failed, see [error code](#) for details.

➤ Dependence

- Header: mi_nir.h

- Library: libmi_nir.a / libmi_nir.so

※ Note

- Currently only DevId =0 is supported;

➤ Example

MI_NIR initialization and de-initialization

```
MI_NIR_DEV  NIRDevId =0;
MI_NIR_CHANNEL  NIRChnId =0;
MI_NIR_DevAttr_t  stDevAttr = {0};
MI_NIR_ChnAttr_t  stChnAttr = {};

stChnAttr.eMode= E_MI_NIR_MODE_NORMAL;

MI_NIR_CreateDevice(NIRDevId, &stDevAttr);
MI_NIR_CreateChannel(NIRDevId, NIRChnId, &stChnAttr);
MI_NIR_StartChannel(NIRDevId, NIRChnId);

/*destroy channel*/
MI_NIR_StopChannel(NIRDevId, NIRChnId);
MI_NIR_DestroyChannel(NIRDevId,NIRChnId);
MI_NIR_DestroyDevice(NIRDevId)
```

➤ Related APIs

MI_NIR_DestroyDevice

2.2. MI_NIR_DestroyDevice

➤ Description

Destroy a NIR device.

➤ Syntax

MI_S32 MI_NIR_DestroyDevice(MI_NIR_DEV DevId);

➤ Parameter

Parameter name	Description	Input/Output
DevId	Device ID	Input

➤ Return value

- MI_SUCCESS(0) successful.
- Non-Zero: failed, see [error code](#) for details.

➤ Dependence

- Header: mi_nir.h
- Library: libmi_nir.a / libmi_nir.so

- ※ Note
 - Currently only DevId = 0 is Supported;

- Example
See example of [MI_NIR_CreateDevice](#)
- Related APIs
MI_NIR_CreateDevice

2.3. MI_NIR_CreateChannel

- Description
Create a NIR channel.
- Syntax
MI_S32 MI_NIR_CreateChannel(MI_NIR_DEV DevId , MI_NIR_CHANNEL ChnId, [MI_NIR_ChannelAttr_t](#) *pstChAttr);

- Parameter

Parameter name	Description	Input/Output
DevId	NIR Device ID	Input
ChnId	Channel ID	Input
pstChAttr	Channel attribute	Input

- Return value
 - MI_SUCCESS(0) successful.
 - Non-Zero: failed, see [error code](#) for details.
- Dependence
 - Header: mi_nir.h
 - Library: libmi_nir.a / libmi_nir.so
- ※ Note
 - Currently only ChnId = 0 is Supported;
 - Create Device firstly, and then create channel
- Example
See example of [MI_NIR_CreateDevice](#)
- Related APIs
MI_NIR_DestroyChannel

2.4. MI_NIR_DestroyChannel

- Description
Destroy a NIR Channel.

➤ Syntax

```
MI_S32 MI_NIR_DestroyChannel(MI_NIR_DEV DevId , MI_NIR_CHANNEL ChnId);
```

➤ Parameter

Parameter name	Description	Input/Output
DevId	Device ID	Input
ChnId	Channel ID	Input

➤ Return value

- MI_SUCCESS(0) successful.
- Non-Zero: failed, see [error code](#) for details.

➤ Dependence

- Header: mi_nir.h
- Library: libmi_nir.a / libmi_nir.so

※ Note

- Currently only ChnId = 0 is supported
- Create Device firstly, and then Create channel

➤ Example

See example of [MI_NIR_CreateDevice](#)

➤ Related APIs

MI_NIR_CreateChannel

2.5. MI_NIR_SetChnParam

➤ Description

Set NIR Channel Parameter.

➤ Syntax

```
MI_S32 MI_NIR_SetChnParam(MI_NIR_DEV DevId , MI_NIR_CHANNEL ChnId, MI_NIR_ChnParam_t *pstChnParam);
```

➤ Parameter

Parameter name	Description	Input/Output
DevId	Device ID	Input
ChnId	Channel ID	Input
pstChnParam	dynamic channel parameters	Input

➤ Return value

- MI_SUCCESS(0) successful.
- Non-Zero: failed, see [error code](#) for details.

➤ Dependence

- Header: mi_nir.h
- Library: libmi_nir.a / libmi_nir.so
- Note
 - Currently only ChnId = 0 is Supported
 - Create Device firstly, and then Create channel

➤ Example

```
MI_NIR_DEV  NIRDevId =0;
MI_NIR_CHANNEL  NIRChnId =0;

MI_NIR_ChnParam_t  stChnParam = {};

MI_NIR_GetChnParam(NIRDevId , NIRChnId,& stChnParam);

stChnParam.eMode= E_MI_NIR_MODE_NORMAL;

MI_NIR_SetChnParam(NIRDevId , NIRChnId, & stChnParam);
```

- Related APIs
- MI_NIR_GetChnParam

2.6. MI_NIR_GetChnParam

- Description
- Get NIR Channel Parameter.
- Syntax
- ```
MI_S32 MI_NIR_GetChnParam(MI_NIR_DEV DevId , MI_NIR_CHANNEL ChnId, MI_NIR_ChnParam
*pstChnParam);
```

➤ Parameter

| Parameter name | Description                | Input/Output |
|----------------|----------------------------|--------------|
| DevId          | Device ID                  | Input        |
| ChnId          | Channel ID                 | Input        |
| pstChnParam    | dynamic channel parameters | Output       |

- Return value
- MI\_SUCCESS(0) successful.
  - Non-Zero: failed, see [error code](#) for details.

- Dependence
- Header: mi\_nir.h
  - Library: libmi\_nir.a / libmi\_nir.so

- ※ Note
- N/A

- Example  
See example of [MI\\_NIR\\_SetChnParam](#)
- Related APIs  
MI\_NIR\_SetChnParam

## 2.7. MI\_NIR\_StartChannel

- Description  
Start NIR Channel.
- Syntax  
MI\_S32 MI\_NIR\_StartChannel(MI\_NIR\_DEV DevId , MI\_NIR\_CHANNEL ChnId);

- Parameter

| Parameter name | Description | Input/Output |
|----------------|-------------|--------------|
| DevId          | Device ID   | Input        |
| ChnId          | Channel ID  | Input        |

- Return value
  - MI\_SUCCESS(0) successful.
  - Non-Zero: failed, see [error code](#) for details.
- Dependence
  - Header: mi\_nir.h
  - Library: libmi\_nir.a / libmi\_nir.so
- Note
  - Only ChnId =0 is supported
  - Create Device firstly, and then Create channel
- Example  
See Example of [MI\\_NIR\\_CreateDevice](#)
- Related APIs  
MI\_NIR\_StopChannel

## 2.8. MI\_NIR\_StopChannel

- Description  
Stop NIR channel.
- Syntax  
MI\_S32 MI\_NIR\_StopChannel(MI\_NIR\_DEV DevId , MI\_NIR\_CHANNEL ChnId);
- Parameter

| Parameter name | Description | Input/Output |
|----------------|-------------|--------------|
| DevId          | Device ID   | Input        |
| ChnId          | Channel ID  | Input        |

- Return value
  - MI\_SUCCESS(0) successful.
  - Non-Zero: failed, see [error code](#) for details.

- Dependence
  - Header: mi\_nir.h
  - Library: libmi\_nir.a / libmi\_nir.so

※ Note  
N/A

- Example  
See example of [MI NIR CreateDevice](#).

- Related APIs  
MI\_NIR\_StartChannel

## 3. data type

### 3.1. Data structure list

Video input related data type definition is as following:

|                                   |                                                 |
|-----------------------------------|-------------------------------------------------|
| <a href="#">MI_NIR_DEV</a>        | Define NIR Device type                          |
| <a href="#">MI_NIR_CHANNEL</a>    | Define NIR Channel type                         |
| <a href="#">MI_NIR_DevAttr_t</a>  | Define NIR Device attribute parameter           |
| <a href="#">MI_NIR_Mode_e</a>     | Define NIR Mode setting                         |
| <a href="#">MI_NIR_ChnAttr_t</a>  | Define NIR Channel static attribute parameter.  |
| <a href="#">MI_NIR_ChnParam_t</a> | Define NIR Channel dynamic attribute parameter. |

### 3.2. MI\_NIR\_DEV

- Description  
Define NIR Device type.
- Definition  
typedef MI\_U32 MI\_NIR\_DEV

## ※ Note

N/A

## ➤ Related data type and interface

N/A

### 3.3. MI\_NIR\_CHANNEL

## ➤ Description

Define NIR Channel type.

## ➤ Definition

typedef MI\_U32 MI\_NIR\_CHANNEL

## ※ Note

N/A

## ➤ Related data type and interface

N/A

### 3.4. MI\_NIR\_DevAttr\_t

## ➤ Description

Define NIR Device attribute parameter.

## ➤ Definition

```
typedef struct MI_NIR_DevAttr_s
{
 MI_U32 u32Reserved;
} MI_NIR_DevAttr_t;
```

## ➤ Member

| Member name | Description   |
|-------------|---------------|
| u32Reserved | Reserved data |

## ※ Note

N/A

## ➤ Related data type and interface

N/A

### 3.5. MI\_NIR\_Mode\_e

## ➤ Description

Define NIR Mode setting, which indicates the method of data process.

## ➤ Definition

```
typedef enum
{
 E_MI_NIR_MODE_NORMAL, //normal
 E_MI_NIR_MODE_BYPASS_VIS, //bypass visible data
 E_MI_NIR_MODE_BYPASS_NIR, //bypass near-infrared data
 E_MI_NIR_MODE_INVALID,
} MI_NIR_Mode_e;
```

※ Note  
N/A

➤ Related data type and interface  
MI\_NIR\_ChnParam\_t;

### 3.6. MI\_NIR\_ChnAttr\_t

➤ Description  
Define NIR Channel static attribute parameter

➤ Definition

```
typedef struct MI_NIR_ChnAttr_s
{
 MI_NIR_Mode_e eMode;
} MI_NIR_ChnAttr_t;
```

➤ Member

| Member name | Description      |
|-------------|------------------|
| eMode       | NIR mode setting |

※ Note  
N/A

➤ Related data type and interface  
N/A

### 3.7. MI\_NIR\_ChnParam\_t

➤ Description  
Define Dynamic parameters of NIR Channel

## ➤ Definition

```
typedef struct MI_NIR_ChnParam_s
{
 MI_NIR_Mode_e eMode;
} MI_NIR_ChnParam_t;
```

## ➤ Member

| Member name | Description      |
|-------------|------------------|
| eMode       | NIR mode setting |

## ※ Note

N/A

## ➤ Related data type and interface

N/A

## 4. NIR ERROR CODE

API error codes are as the table shown below:

| Error code | Macro definition        | Description                           |
|------------|-------------------------|---------------------------------------|
| 0          | MI_NIR_RET_SUCCESS      | Success                               |
| 0xA0272001 | MI_ERR_NIR_MOD_NOT_INIT | Module is not initialized             |
| 0xA0272002 | MI_ERR_NIR_DEV_INVALID  | Invalid device ID                     |
| 0xA0272003 | MI_ERR_NIR_NORESOURCE   | Device does not have enough resources |
| 0xA0272004 | MI_ERR_NIR_CHN_INVALID  | Invalid channel ID                    |
| 0xA0272005 | MI_ERR_NIR_TIMEOUT      | NIR Device Running Timeout            |
| 0xA0272006 | MI_ERR_NIR_NOMEM        | Failed to allocate memory             |
| 0xA0272007 | MI_ERR_NIR_NOTSUPPORT   | Not Supported                         |
| 0xA0272008 | MI_ERR_NIR_BUSY         | Channel system is busy                |
| 0xA0272009 | MI_ERR_NIR_NOTREADY     | Device is not initialized             |
| 0xA027200A | MI_ERR_NIR_NULL_PTR     | Null pointer parameter                |